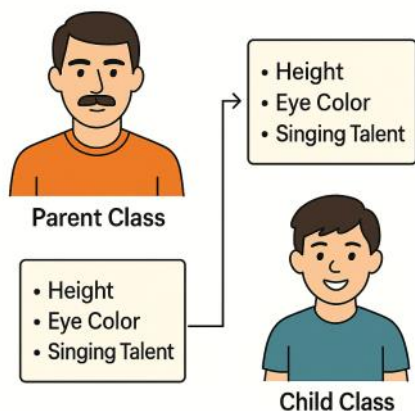


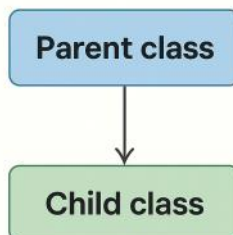
Understanding Inheritance with a Real-World Analogy



Inheritance means one class (child) gets the traits and behaviors of another class (parent), just like a child inherits features from their father.

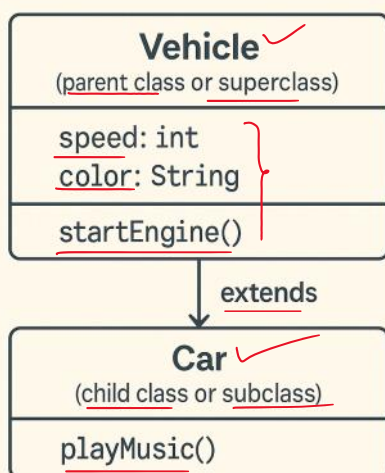
What is Inheritance?

In Java, Inheritance is a mechanism that allows one class to acquire the properties and behavior of another class.



We call the existing class the parent class or superclass, and the new class that inherits from it is called child class or subclass.

Inheritance



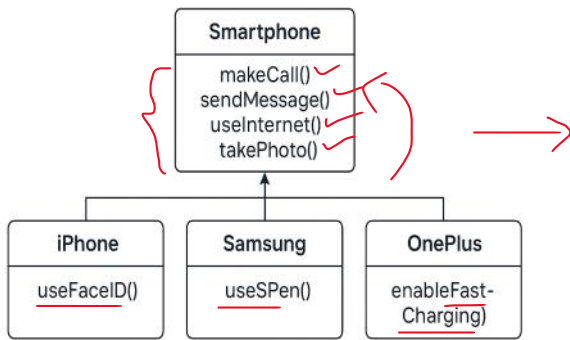
```
// Parent class
class Vehicle {
    int speed = 60;
    String color = "Red";

    void startEngine() {
        System.out.println("Engine started");
    }
}

// Child class
class Car extends Vehicle {
    void playMusic() {
        System.out.println("Playing music");
    }
}

// Main class
public class Main {
    public static void main(String[] args) {
        Car myCar = new Car();
        myCar.startEngine(); // Inherited method
        myCar.playMusic();   // Unique to Car
        System.out.println("Speed: " + myCar.speed); // Inherited field
        System.out.println("Color: " + myCar.color); // Inherited field
    }
}
```

Inheritance Example – Smartphone Brands



Each brand inherits common features from Smartphone and adds unique features – just like Java classes use inheritance for reuse and extension.

```
class Smartphone {
    // Method 1: make a phone call
    public void makeCall() {
        System.out.println("Calling from Smartphone...");
    }

    // Method 2: send a message
    public void sendMessage() {
        System.out.println("Sending message from Smartphone...");
    }
}

// This is the child class
class iPhone extends Smartphone {
    // New method specific to iPhone
    public void useFaceID() {
        System.out.println("Unlocking iPhone with Face ID...");
    }
}

// Main class to run the code
public class Main {
    public static void main(String[] args) {
        // Create object of iPhone
        iPhone myPhone = new iPhone();

        // Call methods from parent class
        myPhone.makeCall();    // inherited
        myPhone.sendMessage(); // inherited

        // Call method from child class
        myPhone.useFaceID();   // specific to iPhone
    }
}
```